Donimoes: A Puzzle with Dominoes

A puzzle where you collect matching dominoes.

Goal

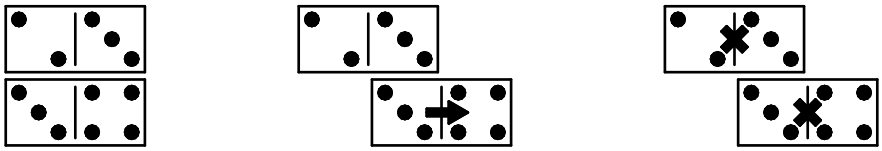
The goal is to collect all the dominoes by sliding matching numbers next to each other.

Moves

There are only two ways a domino can move.

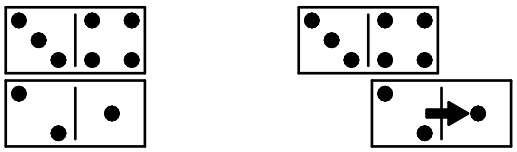
Matching

Move a domino one space along its long axis so that it ends up with at least one of its numbers matching an adjacent number on a neighbouring domino. Then collect the domino you moved and any dominoes that match it, by removing them from the pattern. In this example, the threes match, so you collect both dominoes: solution found!

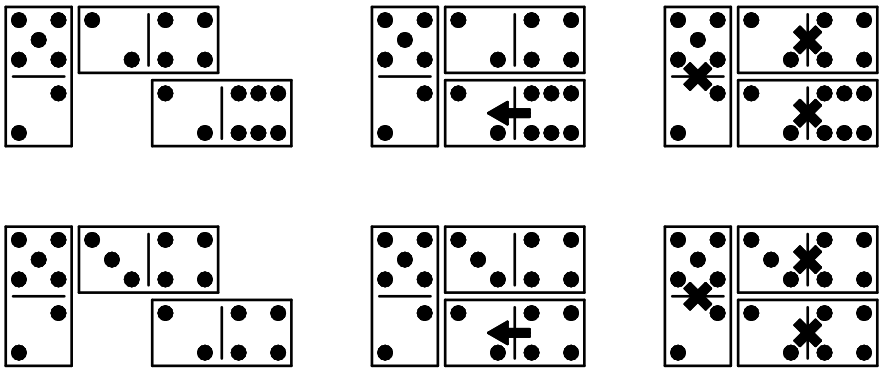


Adding

Move a domino one space along its long axis so that it ends up with at least one of its numbers next to an adjacent number that adds up to six. With an adding move, no dominoes are removed. In this example, the two adds up with the four above it to make six.



Sometimes, you can collect more than two dominoes at once. In the first example, the two matches twos on both of the other dominoes, and you collect all three dominoes. In the second example, the two matches the two to the left, and the four matches the four above it. You collect all three dominoes.



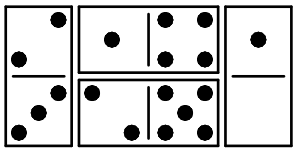
Stay Connected

All the dominoes must stay in one connected group, you can't split the group after moving or after removing the matching dominoes.

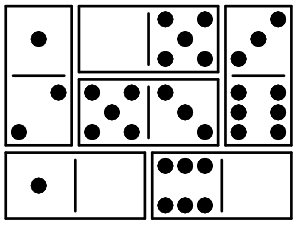
Problems

Here are the starting positions for several Donimoes problems. The solutions are listed at the end.

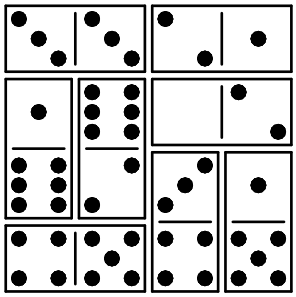
Problem 1



Problem 2



Problem 3



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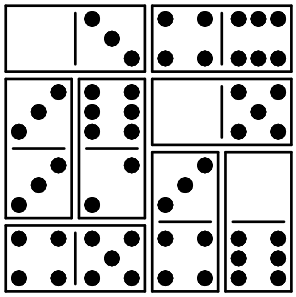
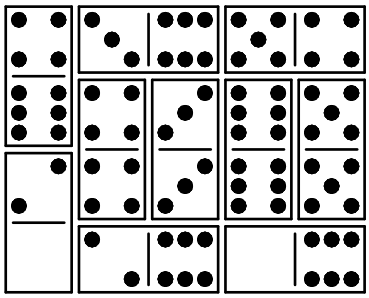


Problem 5



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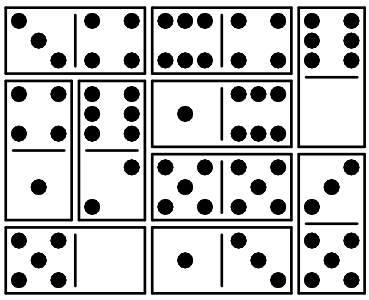

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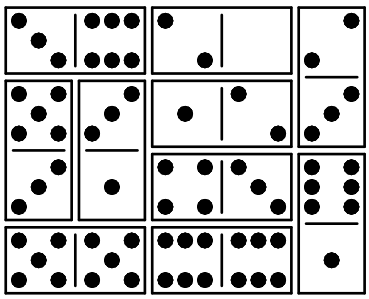
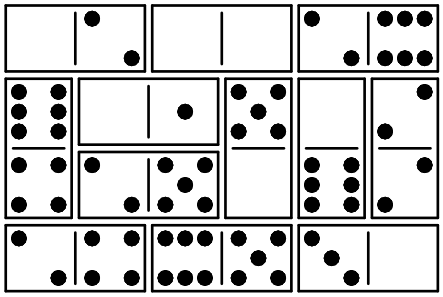


Problem 8



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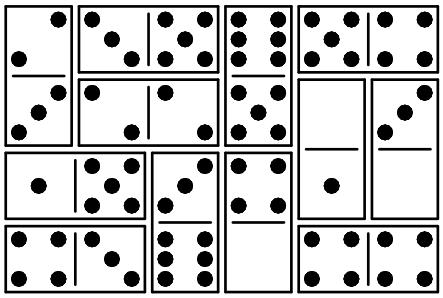

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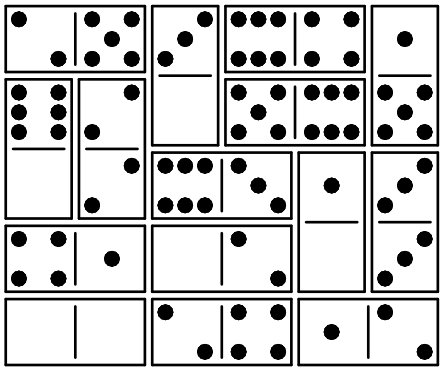

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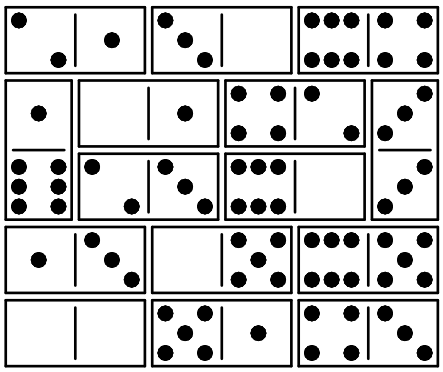

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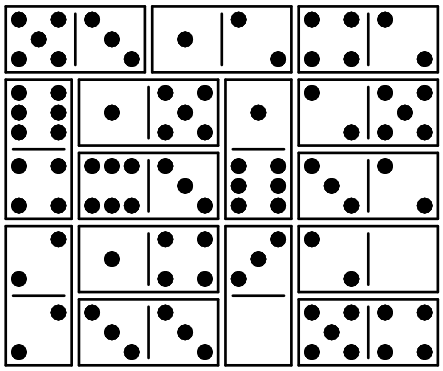

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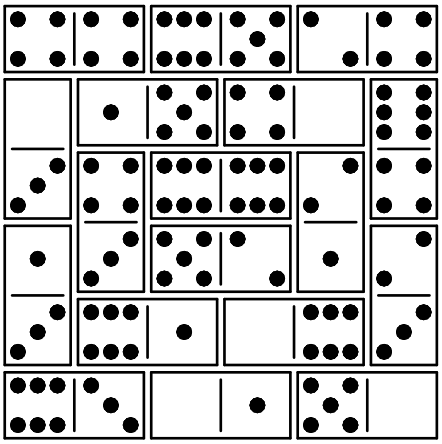
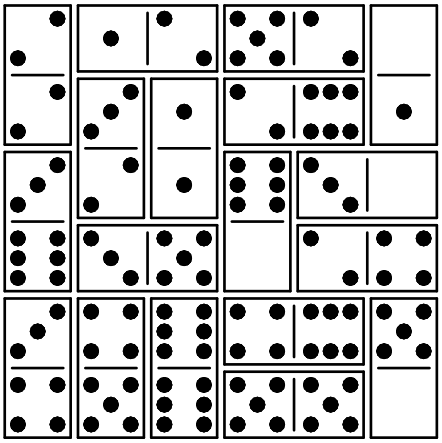
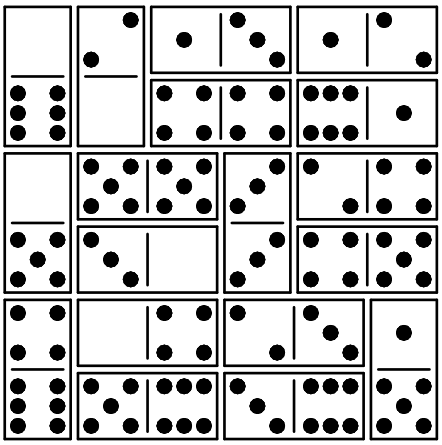


Problem 14



Problem 15



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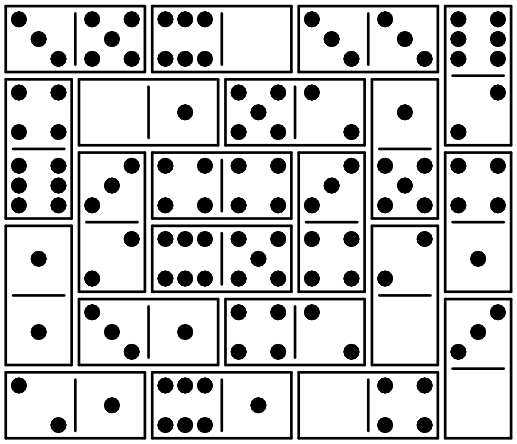
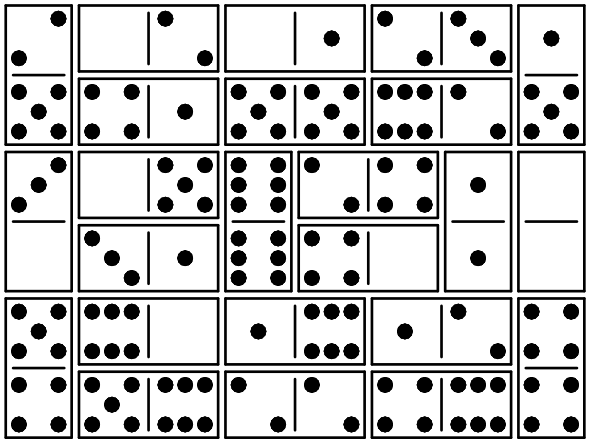


Problem 17



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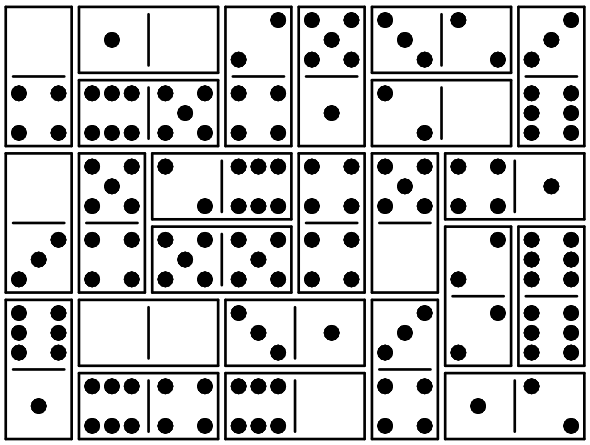

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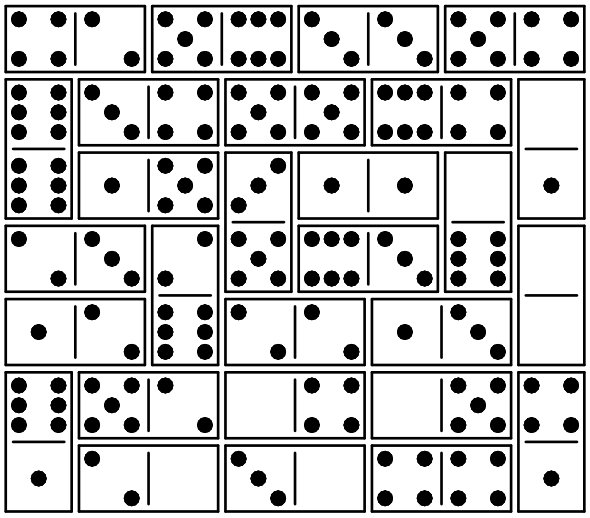
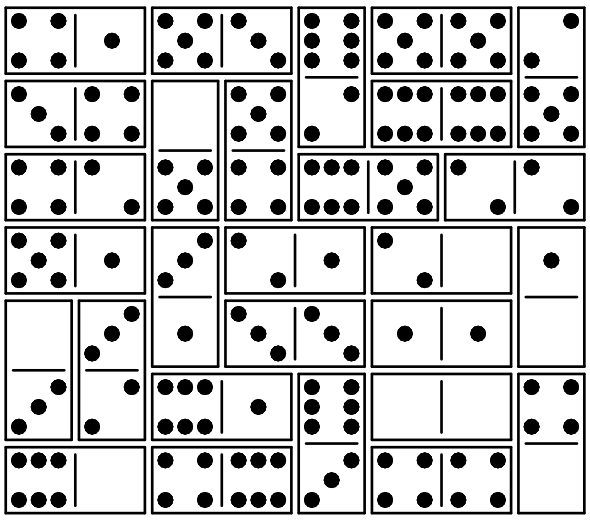


Problem 20



Contributing

Found some interesting problems to solve? Ideas to share? Get in touch at donkirkby.github.com/donimoes.

Solutions

Here are the solutions. For each step, move the listed domino left, right, up, or down. Then make captures for any matching numbers.

1. 10D, 14R, 23D, 14R
2. 60R, 10R, 12D, 12D, 53R
3. 21R, 34U, 45R, 34U, 16U
4. 03L, 46L, 05R, 34U, 45R, 46L, 45R
5. 54R, 20D, 46D, 06L, 36R
6. 34L, 64L, 55L, 35D, 50R, 60D
7. 61D, 43R, 55R, 53D, 53D, 20R, 36R
8. 26R, 24L, 65L, 01L, 30L, 06U, 30L
9. 40D, 65D, 54L, 01D, 36D, 15R, 23D

10. 30U, 56L, 30D, 10U, 10U, 63R, 30D, 41R, 60D, 25R, 41R, 41R
11. 21L, 23L, 60L, 05L, 65L, 33D, 42R, 42R, 01R, 30L, 60R
12. 30D, 54L, 20L, 54L, 54L, 20L, 20L, 64D, 25R, 15L, 15L, 12R
13. 63L, 01L, 50L, 50L, 21D, 66R, 64D, 40R, 40R, 15R, 43U
14. 50D, 45D, 36D, 22D, 66U, 46R, 46R, 60D, 30L, 01D, 01D, 26R, 12R
15. 12R, 44R, 44R, 33U, 20U, 05U, 46U, 56L, 56L, 30L, 45L, 15U, 36R, 15U
16. 21L, 61L, 04L, 04L, 20D, 15D, 52R, 01R, 32U, 46D, 65L, 41U, 34U, 01R, 35R
17. 54D, 60L, 60L, 25D, 02L, 02L, 01L, 01L, 55L, 62L, 11U, 11U, 40R, 44U, 46R, 22R, 46R, 22R, 16L
18. 61D, 60L, 60L, 31L, 31L, 03D, 04D, 10L, 54D, 26L, 10L, 26L, 24D, 51D, 51D, 41L, 36D, 41R, 34U, 34U, 12L, 20R, 34U
19. 41D, 30R, 12L, 61U, 61U, 20R, 26D, 13L, 34L, 06D, 34L, 55L, 35U, 35U, 11R, 64R
20. 25U, 40D, 10D, 63D, 60R, 60R, 32D, 20R, 31D, 65R, 62D, 21R, 51R, 34L, 53R, 51R

Donimoes is an original puzzle designed by Don Kirkby.